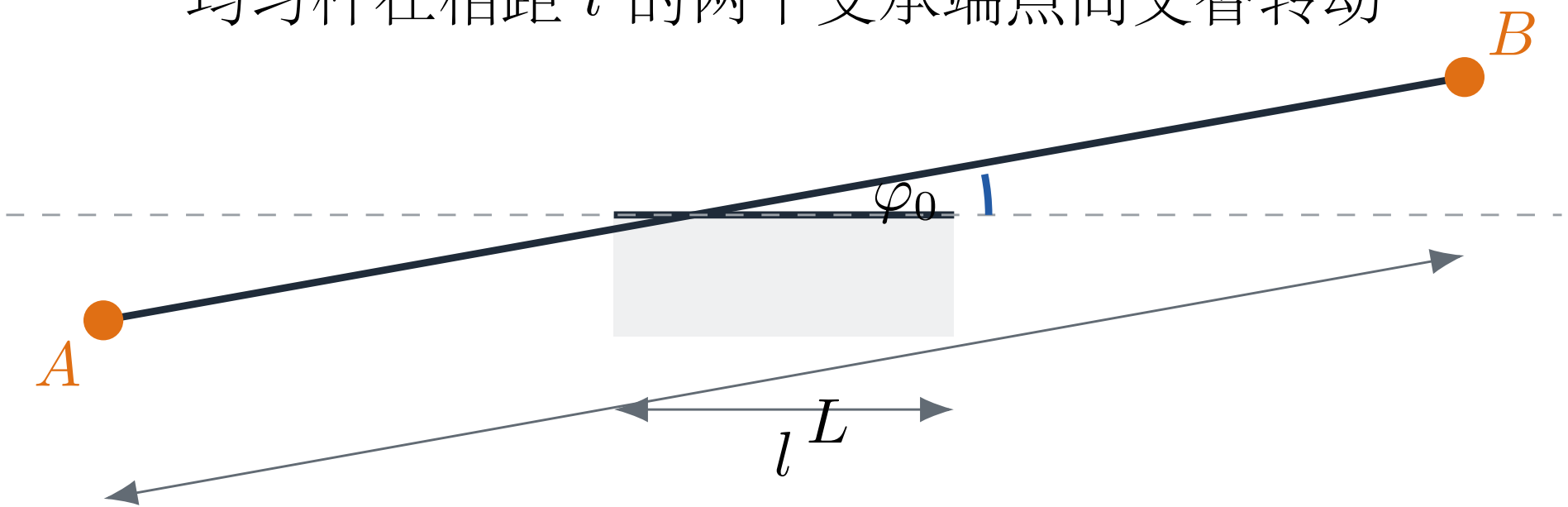
均匀杆在相距 l 的两个支承端点间交替转动

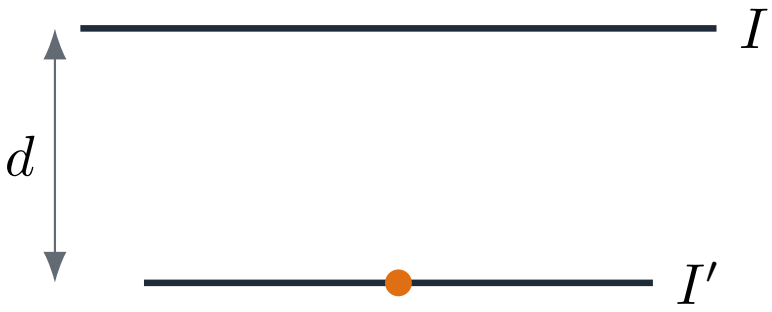


$$|\ddot{\varphi}| = \frac{6gl}{L^2}, \qquad t_1 = L\sqrt{\frac{\varphi_0}{3gl}}, \qquad T = 4t_1$$

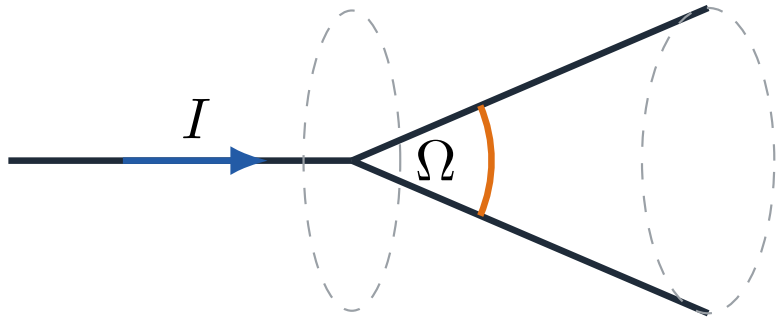


初始极限位置　　到达水平位置　　关于右端上摆　　返回水平位置

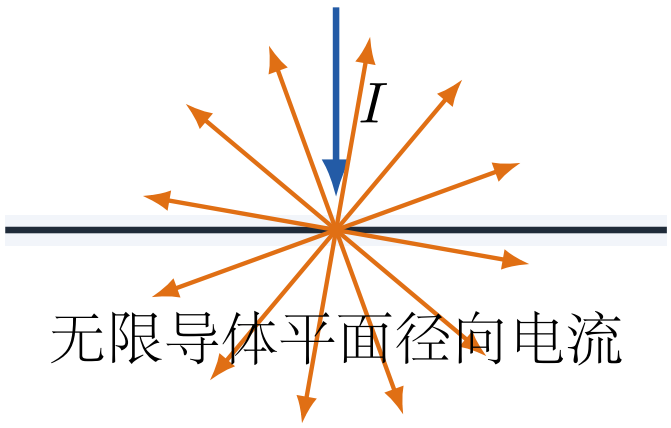
完全弹性碰撞使运动在两个支承端点之间镜像延续



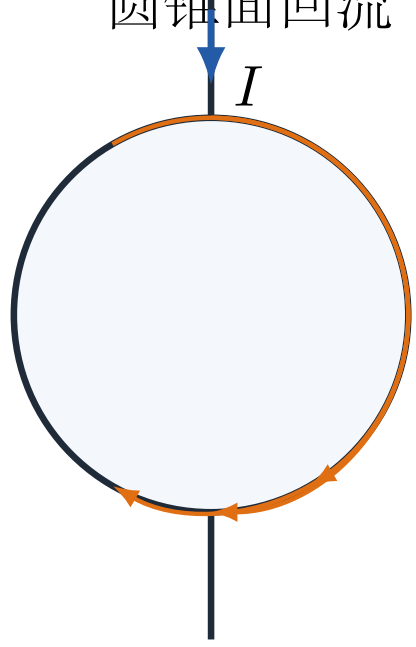
可转动载流铁棒



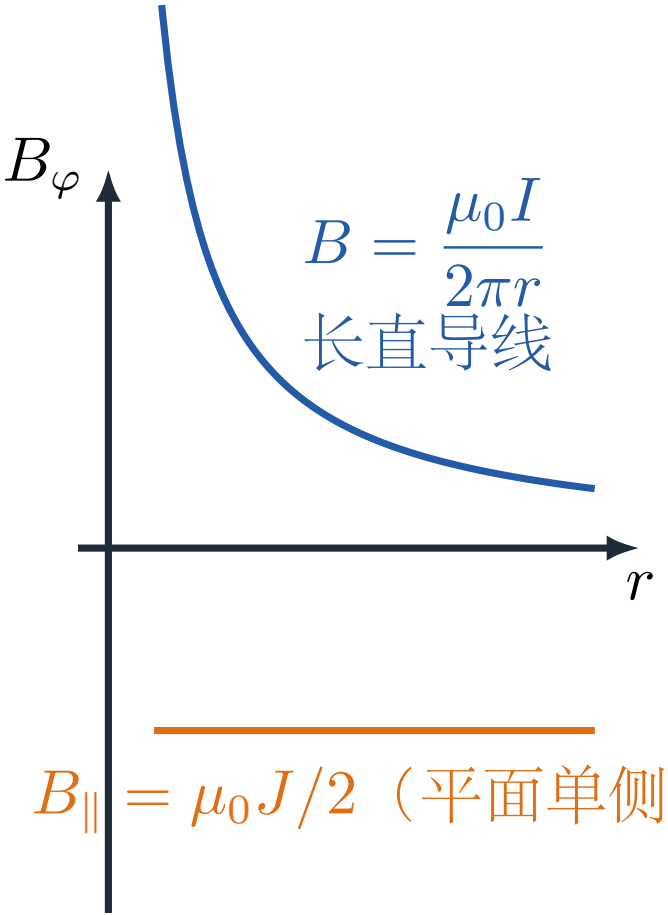
圆锥面回流



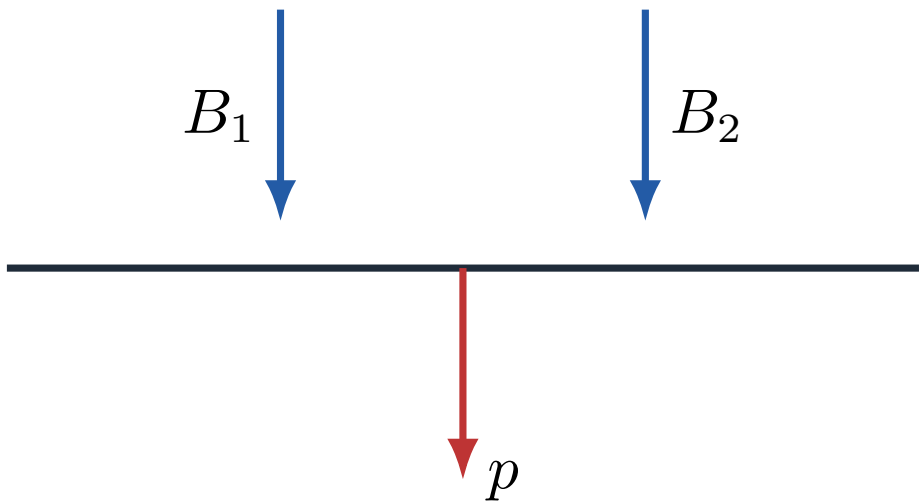
无限导体平面径向电流



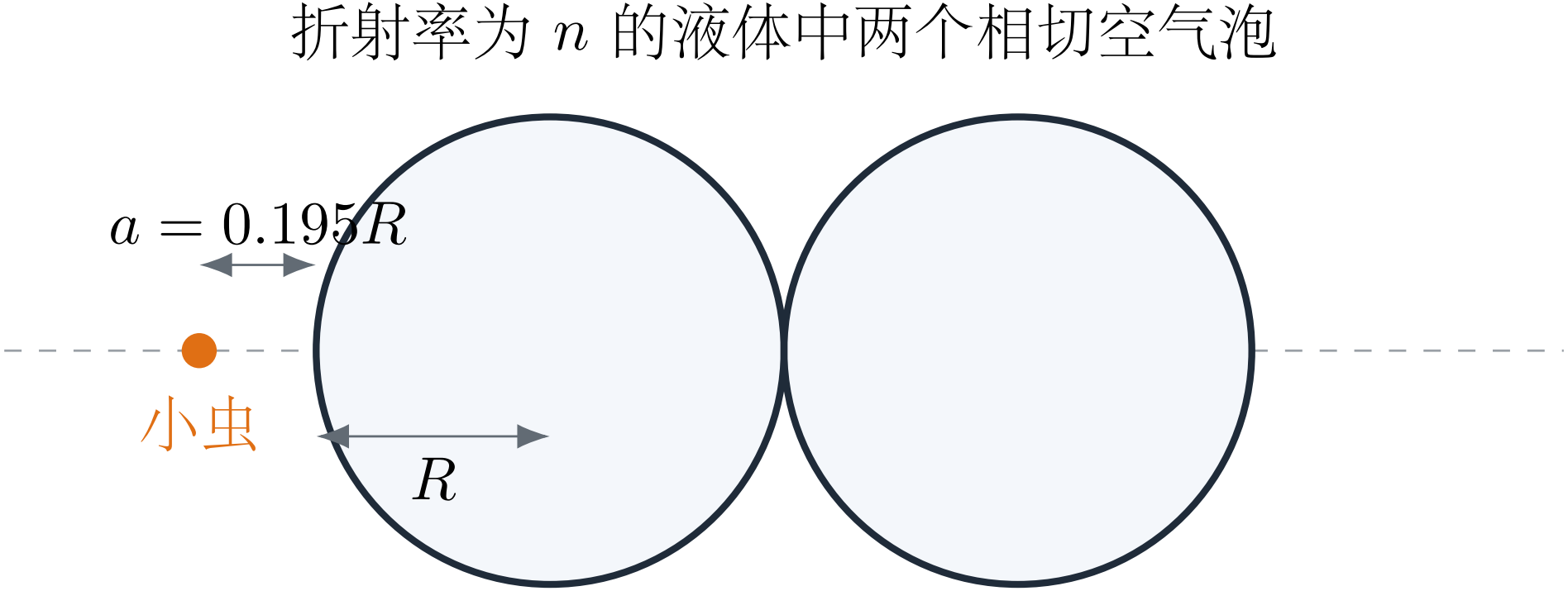
球壳表面电流

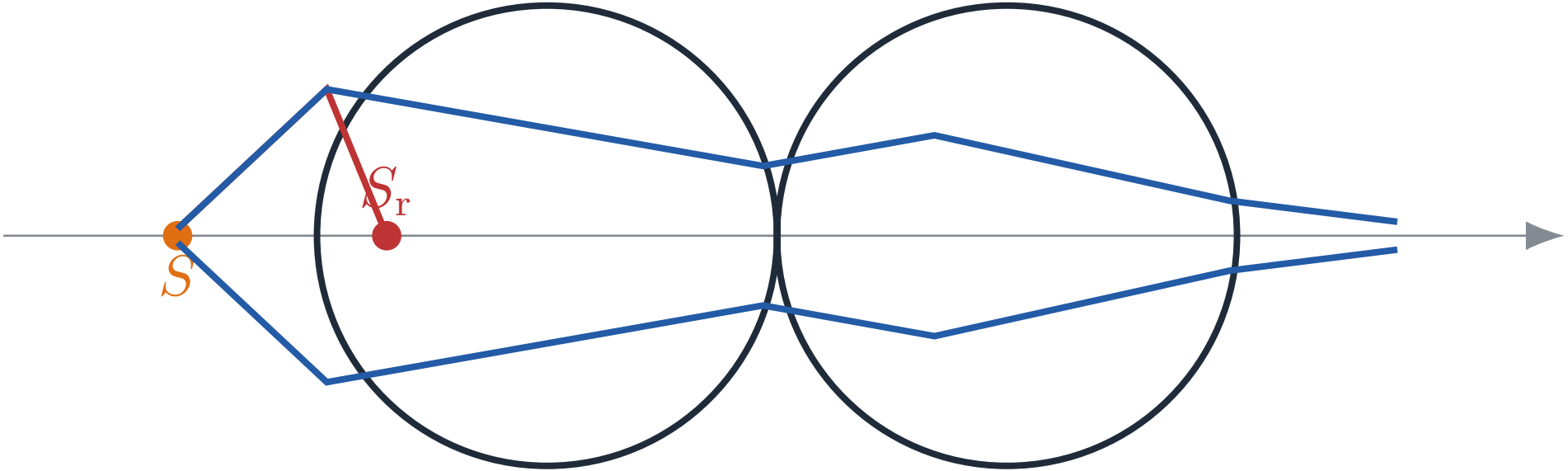


磁场边界条件与磁压

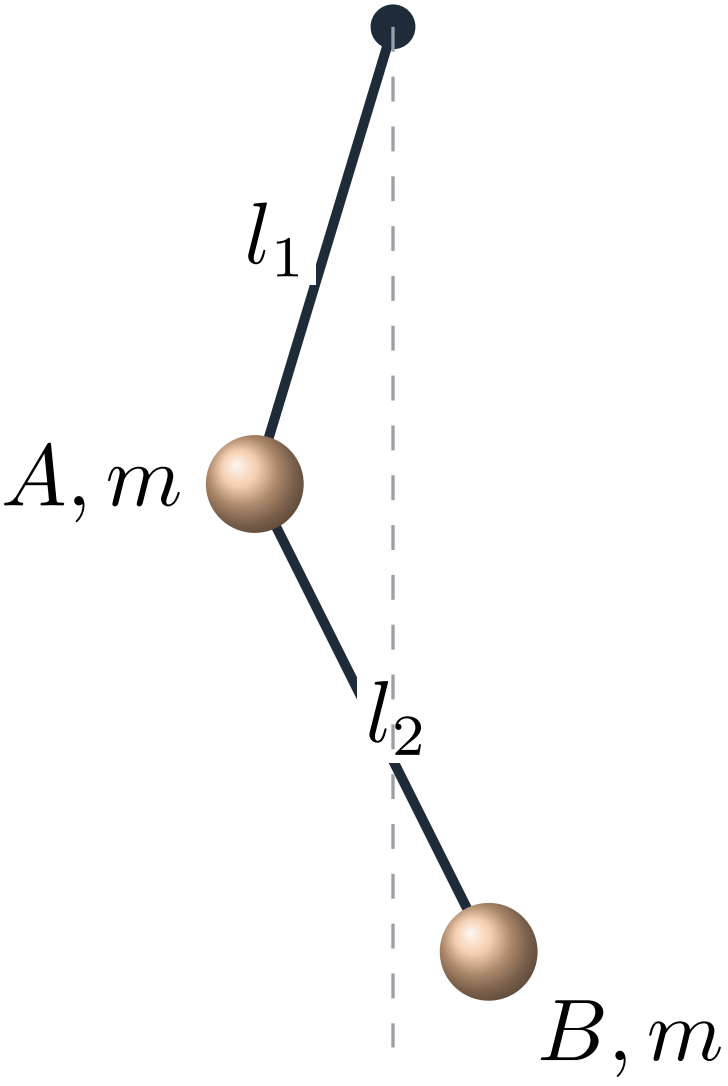


$$J = \frac{B_1 - B_2}{\mu_0}, \qquad p = \frac{B_1^2 - B_2^2}{2\mu_0}$$


折射率为 n 的液体中两个相切空气泡
$a = 0.195R$
小虫
R

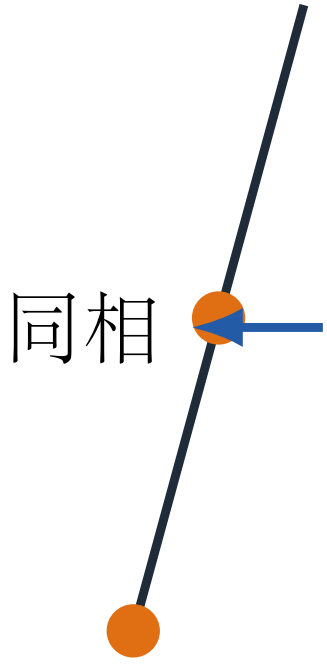


$$\Phi_{\text{bubble}} = -\frac{2(n-1)}{R}, \qquad \text{像重合条件给出 } n \approx 1.3$$

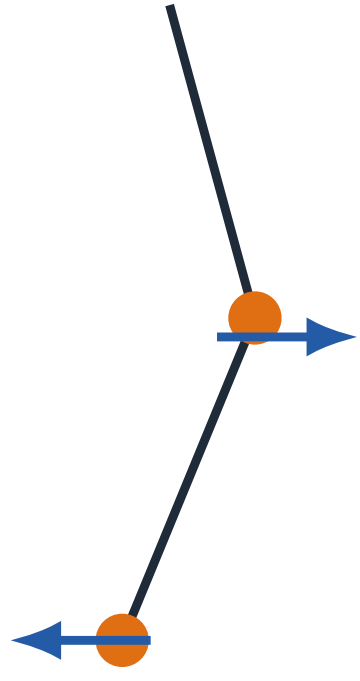


两级等质量复摆，绳长随温度改变

$\alpha_1 + \alpha_2 = 0$ 时两支频率的一阶温漂同时消失



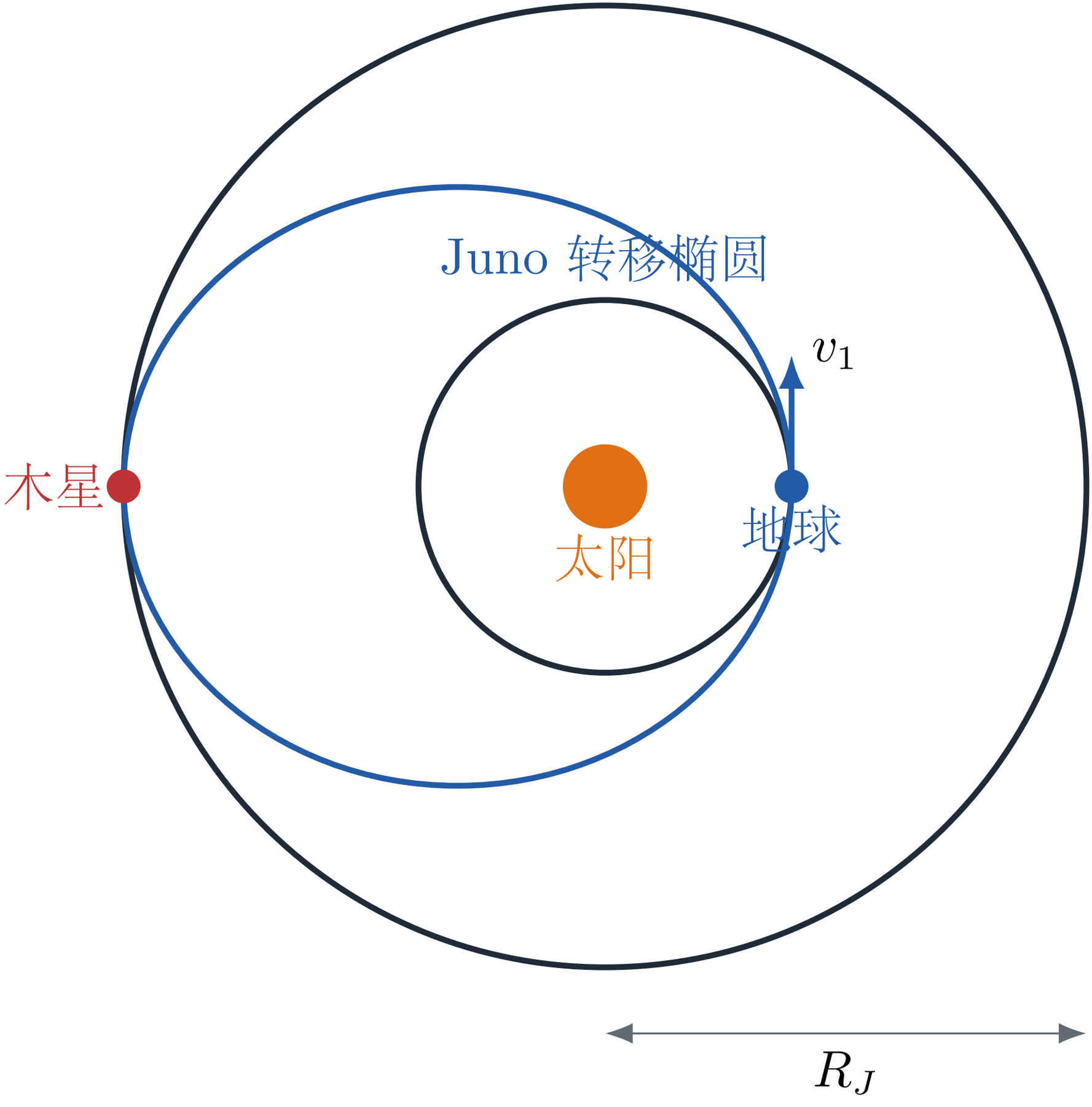
同相



反相

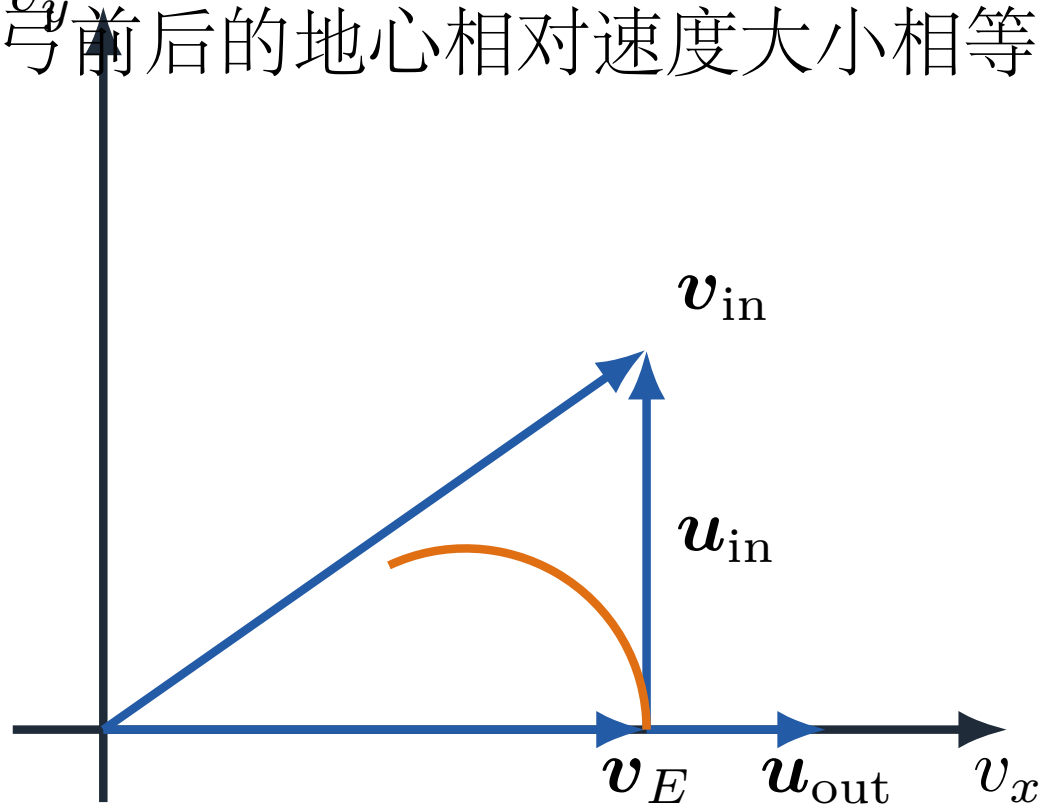
$\omega_-^2 = (2 - \sqrt{2})g/l_0$

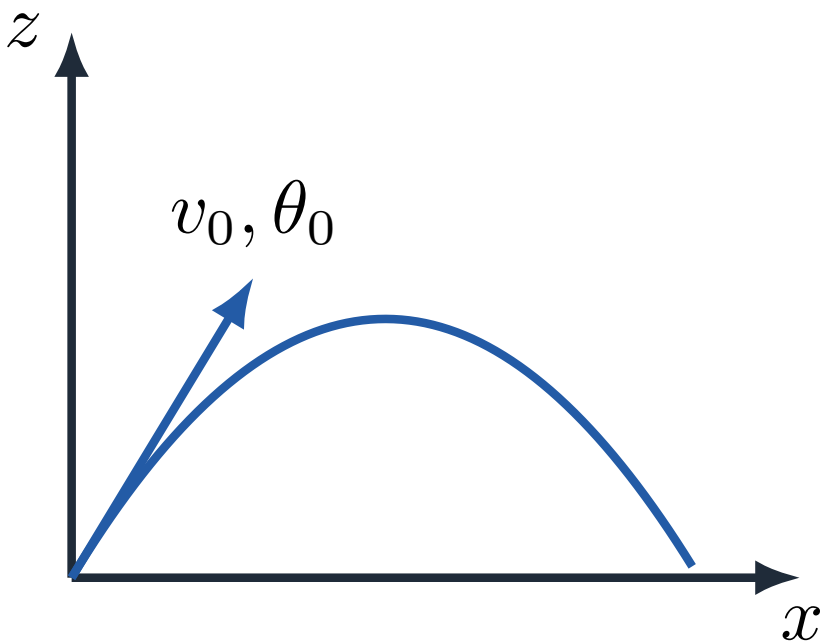
$\omega_+^2 = (2 + \sqrt{2})g/l_0$


Juno 转移椭圆
v_1
木星
太阳
地球
R_J

地球引力弹弓前后的地心相对速度大小相等

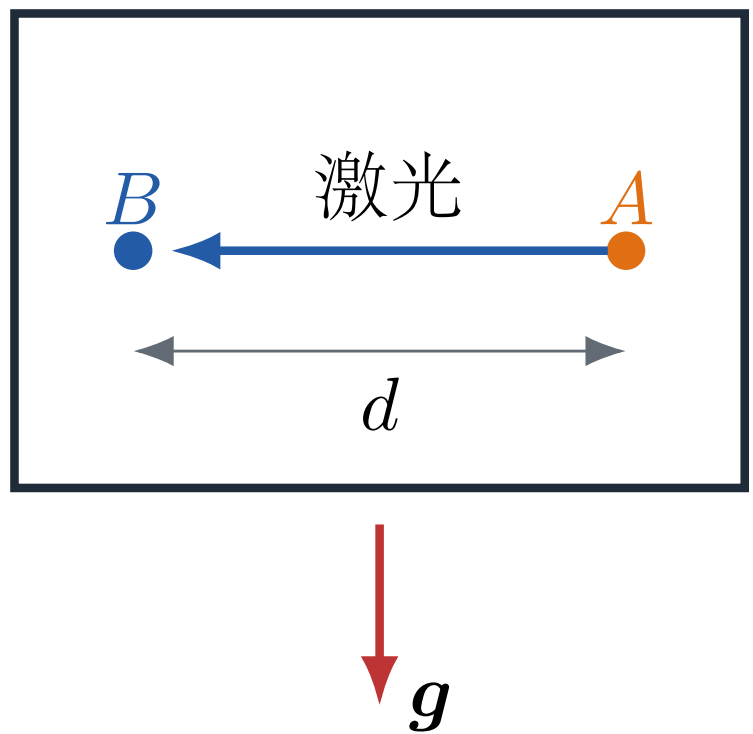
$$R_J/R_E = T_J^{2/3}$$
$$v_p^2 = \mu_\odot(2/R_E - 1/a)$$
$$v_a R_J = v_p R_E$$

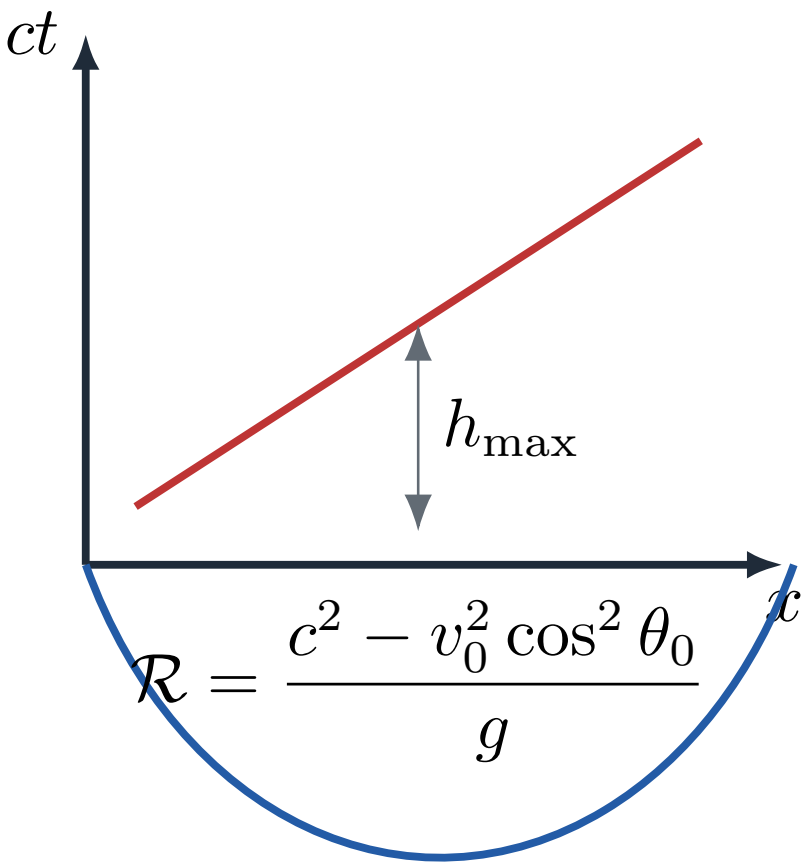


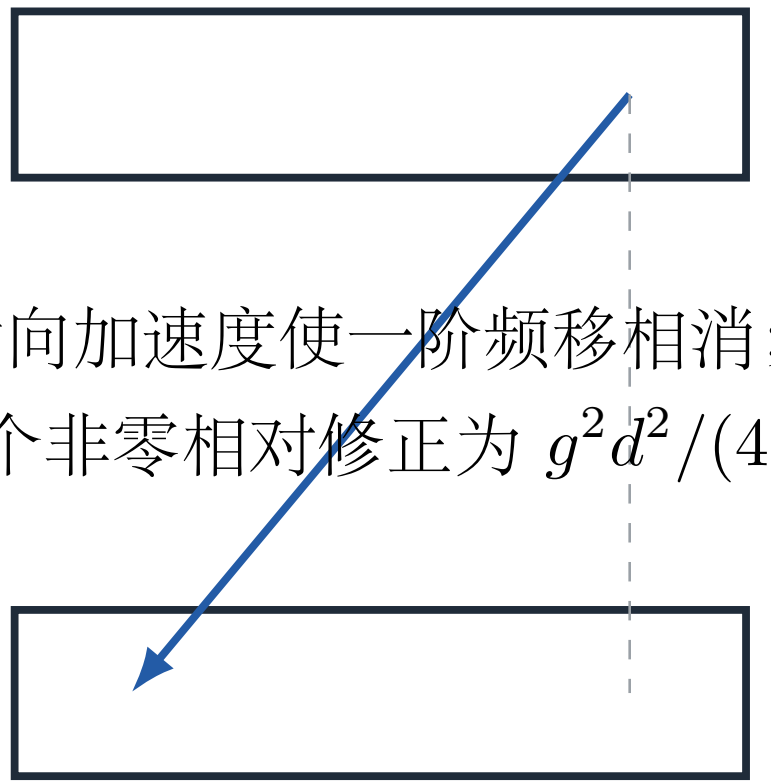


抛体事件轨迹

横向分离的加速飞船

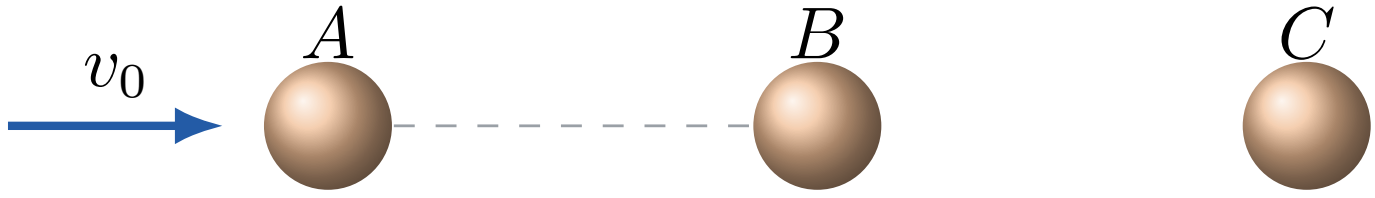




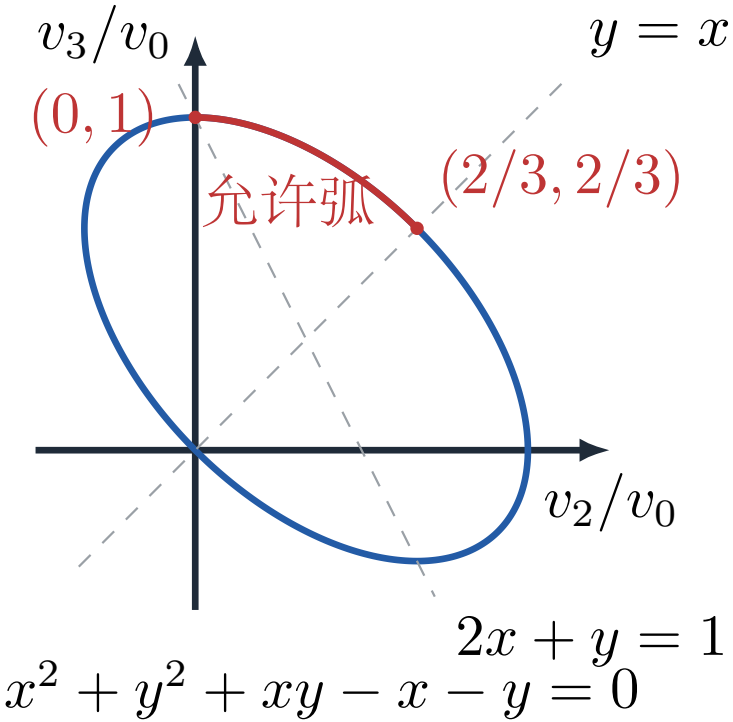


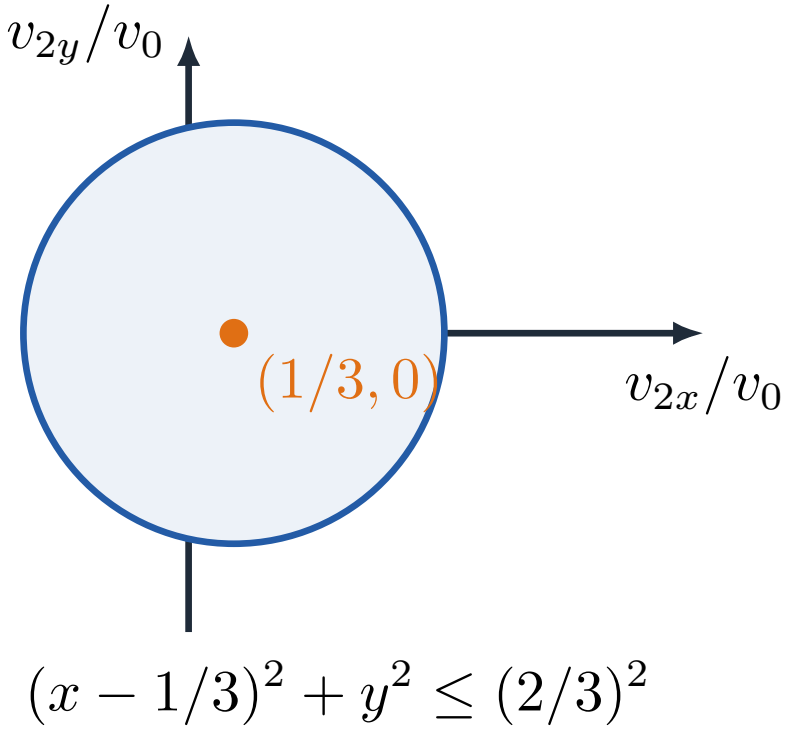
横向加速度使一阶频移相消；

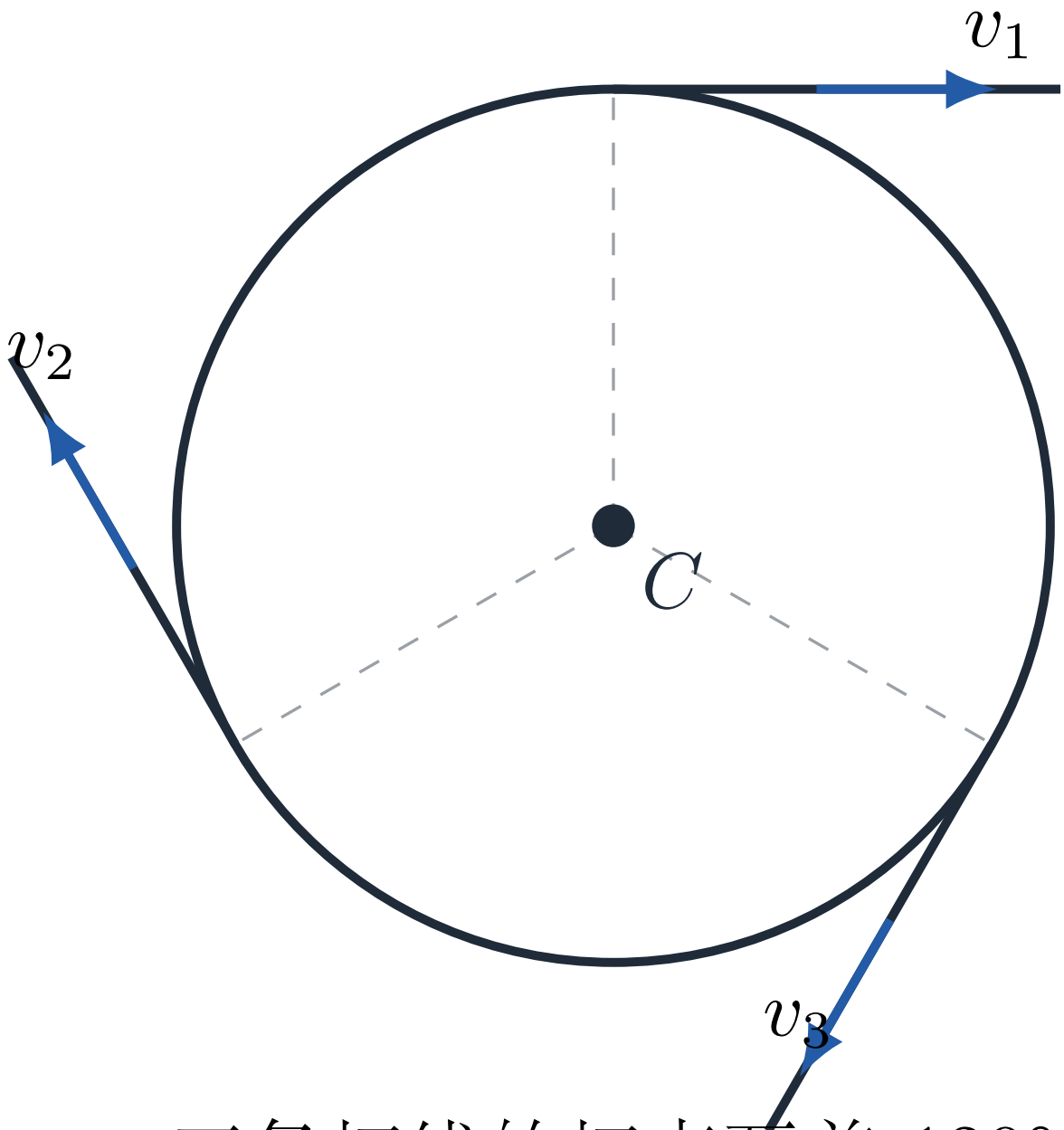
首个非零相对修正为 $g^2 d^2/(4c^4)$



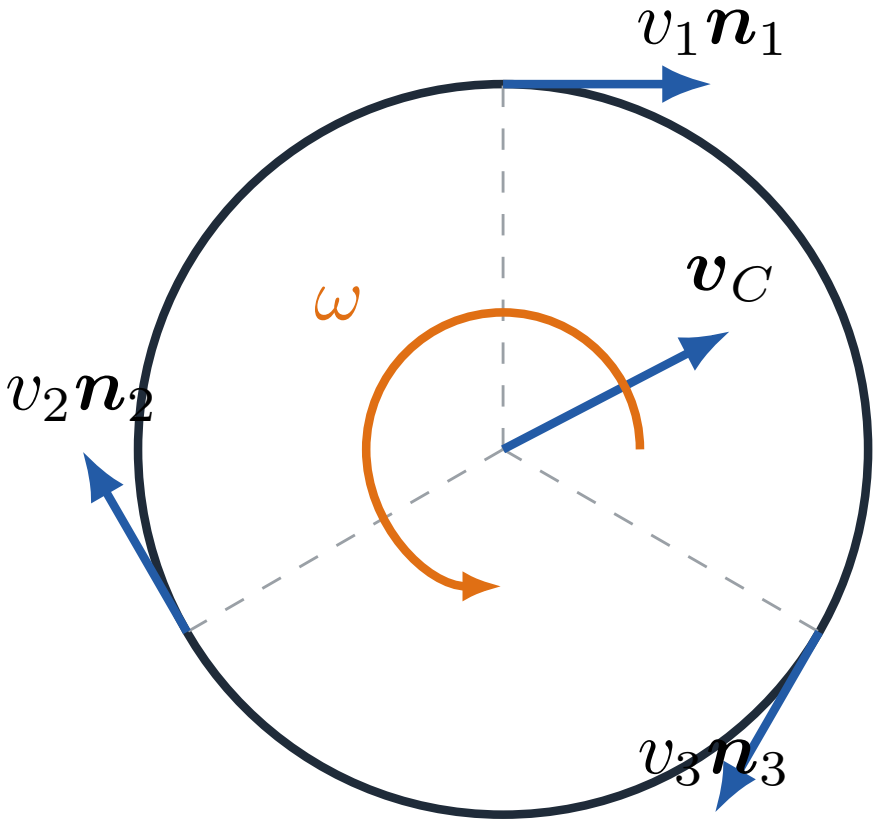
三个等质量小球的一维或非共线弹性碰撞


v_3/v_0
$y = x$
$(0, 1)$
允许弧
$(2/3, 2/3)$
v_2/v_0
$2x + y = 1$
$x^2 + y^2 + xy - x - y = 0$


v_{2y}/v_0
$(1/3, 0)$
v_{2x}/v_0
$(x - 1/3)^2 + y^2 \leq (2/3)^2$



三条切线的切点互差 120°



$$\boldsymbol{v}_C \cdot \boldsymbol{n}_i + \omega R = v_i$$

$$\omega R = \frac{v_1 + v_2 + v_3}{3}$$

$$v_C = \frac{2}{3}\sqrt{\sum v_i^2 - \sum_{i<j} v_i v_j}$$

$$\boldsymbol{a}_C = \boldsymbol{0}, \quad \dot{\omega} = 0$$